

## CELLCHECK · DESIGN DECISIONS REGISTER

# CellCheck design decisions register

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## DOCUMENT CONTROL

|             |                                           |         |                |
|-------------|-------------------------------------------|---------|----------------|
| Document ID | CCK-DEC-001                               | Version | 0.1            |
| Status      | <b>Draft</b>                              | Date    | 2026-10-01     |
| Author      | Amish Chadha                              | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept             |         |                |
| Repository  | github.com/BoujeeEnjinia1701/cellcheck    |         |                |
| Source      | docs/06-design-decisions.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                |
|---------|------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 2026-10-01 | Amish Chadha | Register opened with the open decisions from the review note and decision records, the build plan work and every decision made so far |

## CellCheck design decisions register

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan ( `docs/05-build-plan.md` ) describes the design as it stands and does not list open decisions.

### ■ Open decisions

*Table 1. Decisions still to be made.*

| #  | Decision needed                                                                             | Options                                                                                                                                            | Recommendation                                                                       | Affects in the build                                            | Source                                      |
|----|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------|
| 1  | How the guard is held down (part of the containment arrangement)                            | (a) two knurled thumb screws, as modelled; (b) two toggle latches; (c) a hinge at the back with one latch                                          | (a) for the first prototype; decide for later units after the TRL 4 containment test | Guard flanges, rivet nuts (build plan sections 3.8 and step 11) | CCK-DDR-003, A1                             |
| 2  | Whether the wire slots under the guard are sealed after wiring                              | (a) open with grommet strip, as modelled; (b) filled round the leads with high-temperature silicone                                                | (b)                                                                                  | Wiring step 9                                                   | CCK-DDR-003, A2                             |
| 3  | First trial partner                                                                         | Repair café, e-bike repair shop or battery collection point                                                                                        | None stated                                                                          | Not part of the build; sets the cells used at TRL 4             | CCK-DDR-001, item 10                        |
| 4  | Scheduling rule for the attended day                                                        | (a) start a current stage only if it finishes the same day (12.0 cells per day); (b) pause overnight with the unit off (12.8 cells per day)        | None made; (b) adds an unquantified capacity error                                   | Firmware sketch only                                            | CCK-DDR-001, item 12                        |
| 5  | Responses to the requirements at risk, R11 (mass, 4.91 kg against 5 kg) and R5 (throughput) | For R11: accept and weigh at TRL 4, or a 1.2 mm tray (saves about 0.49 kg but thins the containment); for R5: scheduling rule, more channels later | None made                                                                            | Tray thickness if R11 changes                                   | CCK-DDR-001, item 14; CCK-CAL-001 section 8 |
| 6  | Grade A resistance limit per cell model (60 mΩ default)                                     | Values per cell model from named data sheets                                                                                                       | None; needs data sheets                                                              | Grading table only                                              | CCK-DDR-001, item 16                        |
| 7  | 18650 cells shown in the product renders where the model uses the 21700 envelope            | Accept for the renders, or render 21700 cells                                                                                                      | Accept; the model keeps the 21700 envelope for clearance checks                      | None                                                            | REVIEW 2026-09-26                           |
| 8  | Bins for matched groups A, B and C shown in the renders                                     | (a) render context only; (b) a BOM line for three bins, costed against the budget                                                                  | (a) now; decide (b) with TRL 4 purchasing                                            | None now                                                        | REVIEW 2026-09-26                           |
| 9  | Fan finger guards shown in the renders                                                      | Add to BOM line 10 at TRL 4 and cost them then, or leave out                                                                                       | Add at TRL 4                                                                         | Fans (step 7)                                                   | REVIEW 2026-09-26                           |
| 10 | Guard perforation drawn as slots and 1.0 mm sheet in the renders                            | Accept as appearance, or redraw with round holes and 0.8 mm                                                                                        | Accept; the BOM specification stays as written                                       | None                                                            | REVIEW 2026-09-26                           |
| 11 | Second-life SwapCell pack for PowerBox                                                      |                                                                                                                                                    |                                                                                      | None                                                            | CCK-DDR-001, item 1; CCK-                   |

| # | Decision needed                            | Options                                               | Recommendation                                | Affects in the build | Source             |
|---|--------------------------------------------|-------------------------------------------------------|-----------------------------------------------|----------------------|--------------------|
|   | (low-current storage, about 334 Wh, 165 W) | Agree with the SwapCell and PowerBox projects, or not | Raise with those projects (cross-repo action) |                      | CAL-001 section 10 |

## ■ To confirm when parts are bought

Table 2. Items to check against the parts before building.

| #  | What to confirm                                                                                                                                   | Why it matters                                                                   | Source                |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------|
| 1  | The steel tray is plain steel, about 460 x 300 x 45 mm with 1.5 mm walls, and its flat floor is at least 457 x 297 mm inside                      | The liner, plate and containment estimate assume it                              | CCK-DDR-003, P1       |
| 2  | The rubber feet have M4 studs about 8 mm long                                                                                                     | The stud must pass the 1.5 mm floor and engage the standoff                      | CCK-DDR-003, P1       |
| 3  | The M4 rivet nuts grip 1.5 mm sheet and their heads are under 1 mm thick                                                                          | The guard flange sits round the head                                             | CCK-DDR-003, P4       |
| 4  | The cell holders have two M3 fixing holes in their base and take 18650 and 21700 cells up to 70 mm                                                | Holder screws and the 84 mm holder length                                        | CCK-DDR-003, P3       |
| 5  | The charger modules, current monitor breakouts, controller and converter have M3 mounting holes (or are fixed to a small perfboard that has them) | Each board stands on two nylon standoffs                                         | CCK-DDR-003, P5       |
| 6  | The 2.8 in display module is about 86 x 50 mm with four corner holes                                                                              | Display stand size                                                               | CCK-DDR-003, P6       |
| 7  | The heatsink spine is 6 mm thick, 50 mm tall, and has a fin gap at 20 mm from the left end and every 40 mm after it                               | MOSFET screws go between fins; plate and shroud screws are tapped into the spine | CCK-DDR-003, P7       |
| 8  | The fans have a 50 mm square screw pattern                                                                                                        | Shroud holes                                                                     | CCK-DDR-003, P8       |
| 9  | The DC jack, fuse holder and rocker switch cut-out sizes against the inlet housing (11.2 mm, 12 mm, 12 x 16 mm)                                   | Inlet housing holes                                                              | CCK-DDR-003, P10      |
| 10 | A sourced figure for the heat released by one 21700 cell in thermal runaway, to replace the assumed 100 kJ                                        | Containment estimate before any TRL 4 work                                       | CCK-CAL-001 section 7 |

## ■ Decisions made

Table 3. Decisions made, with dates and records.

| Date       | Decision                                                                                                                                                                                                                                                                                                                                                                                                                               | Decided by                                                                                                                                                                                                                           | Record                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 2026-09-25 | Keep SwapCell in the pitch and plan a second-life variant; charge and discharge attended only (rest stage may be left); hardware watchdog and undervoltage comparators; grading thresholds, 14-day rest and $\pm 1$ % group tolerance as starting values; TP5100, INA226 and ESP32-S3 class modules; eight channels with a linear load; LFP profile later; per-cell CSV mapped to the ReflowEconomy passport (items 1 to 3 and 5 to 9) | Amish: "i accept all your recommendations, go with them across all repos."                                                                                                                                                           | CCK-<br>DDR-001,<br>CCK-<br>DDR-002 |
| 2026-09-25 | Budget raised from \$160 to \$175 (item 4)                                                                                                                                                                                                                                                                                                                                                                                             | Amish, same instruction                                                                                                                                                                                                              | CCK-<br>DDR-001,<br>CCK-<br>DDR-002 |
| 2026-09-25 | DC four-wire resistance and a 14-day rest (item 11); firmware and build rules: same-channel cohort self-discharge reading, temperature-gated resistance pulse, per-channel calibration, fan-fault stop, detached-thermistor check (item 13); raise the passport gaps with ReflowEconomy (item 15)                                                                                                                                      | Amish, same instruction                                                                                                                                                                                                              | CCK-<br>DDR-001,<br>CCK-<br>DDR-002 |
| 2026-10-01 | Design for construction: feet and standoffs, 1.5 mm plate on six standoffs, holder fixings and wire slots, folded guard with thumb screws, board standoffs, larger display stand, heatsink and shroud fixings, thermistor C-clips, inlet housing, 5 V converter and channel boards (P1 to P12)                                                                                                                                         | Made under Amish's 2026-09-30 instruction: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." Open for his review | CCK-<br>DDR-003                     |